

Technical Executive Summary: AI-SaMD Clinical Validation for Thai FDA Registration

Regulatory Affairs & Clinical Affairs Directorate

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Abstract

This Technical Executive Summary is compiled in support of the Thai Food and Drug Administration (Thai FDA) Medical Device Control Division submission for the **RetinaAI-Dx** platform (Class 3 Medical Device, Rule 10). It provides an executive overview of device identification, clinical validation against consensus standard ground truth, and comprehensive compliance with ASEAN Common Submission Dossier Template (CSDT) and international quality benchmarks.

1 Device Identification & Intended Use

- **Proprietary / Trade Name:** RetinaAI-Dx Diagnostic Engine
- **Software Version:** v2.4.1 (Build 2026.1)
- **Device Classification:** Class 3 (ASEAN CSDT / Thai FDA Classification Rule 10 for Active Diagnostic Software)
- **Basic UDI-DI / Identifier:** 885901234RETINA01V5
- **Intended Use:** The RetinaAI-Dx platform is a Software as a Medical Device (SaMD) cloud- and edge-integrated algorithm intended for automated analysis of non-mydriatic digital fundus photographs. It provides diagnostic screening recommendations and detects signs of moderate-or-worse Diabetic Retinopathy (more than mild non-proliferative diabetic retinopathy [NPDR]) and referable Diabetic Macular Edema (DME) in adult diabetic patients attending hospital and primary-care outpatient settings.

2 Summary of Clinical Performance

A multi-center, retrospective-prospective pivotal clinical validation study was conducted across three tertiary hospital centers in Thailand ($N = 4,280$ fundus images from 2,140 patients). The algorithmic diagnostic output was benchmarked against an independent clinical reference standard adjudicated by an expert panel of three Thai board-certified retinal ophthalmologists.

Table 1 compares the clinical performance endpoints against predetermined Thai FDA clinical acceptance criteria. Statistical uncertainty is expressed as two-sided 95% Confidence Intervals (95% CI).

The clinical evaluation demonstrates that the software's receiver operating characteristic area (AUC-ROC = 0.962) surpasses clinical thresholds without introducing racial or demographic algorithmic bias across the Thai test cohort.

Table 1: Clinical Performance Metrics vs. Thai FDA Acceptance Thresholds

Metric	Observed	95% CI	FDA Target	Status
Sensitivity	93.4%	[91.8%, 94.8%]	$\geq 85.0\%$	Met (Superior)
Specificity	91.2%	[89.9%, 92.4%]	$\geq 85.0\%$	Met (Superior)
Positive Predictive Value (PPV)	84.6%	[82.5%, 86.5%]	$\geq 75.0\%$	Met
Negative Predictive Value (NPV)	96.1%	[95.1%, 97.0%]	$\geq 90.0\%$	Met
AUC-ROC	0.962	[0.951, 0.972]	≥ 0.900	Met (Robust)
Image Unreadable Rate	2.1%	[1.5%, 2.8%]	$\leq 5.0\%$	Met

3 Regulatory Standards Compliance

The development, verification, validation, and risk controls for the RetinaAI-Dx platform strictly align with harmonized international standards mandated for Thai FDA Class 3 Medical Device Registration:

- ISO 13485:2016 (Medical Devices – Quality Management Systems):**
 The legal manufacturer maintains an audited QMS covering software design controls, supplier qualification, change management, and post-market clinical follow-up (PMCF) routines.
- IEC 62304:2006/AMD 1:2015 (Medical Device Software – Software Life Cycle Processes):**
 The software architecture is categorized as **Safety Class B**. Software requirements, architectural design, unit testing, integration verification, cybersecurity hardening, and release gates follow an end-to-end traced lifecycle.
- ISO 14971:2019 (Application of Risk Management to Medical Devices):**
 A closed-loop risk management system was executed across all lifecycle stages. Specific SaMD hazards—including ungradable optical inputs, compression artifacts, and cloud API network interruptions—were mapped to hardware/software risk control mitigations, confirming that all individual and overall residual risks are acceptable.

Conclusion: The clinical and technical data compiled in this summary establish that RetinaAI-Dx performs safely and effectively according to its intended use, satisfying the requirements for Thai FDA Class 3 marketing clearance.